

Dynamic capabilities and multi-sided platforms: Fostering organizational agility, flexibility, and resilience in B2B service ecosystems

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ABSTRACT

The use of multi-sided platforms in business-to-business services can redefine how established businesses and their operational activities meet the requirements of volatile business markets. Using an abductive research approach, we analyzed how the dynamic capabilities of service firms as platform owners foster the organizational capabilities of agility, flexibility, and resilience in a multi-sided platform ecosystem. Using data from 34 interviews, and based on the Gioia methodology, we illustrated the underlying mechanisms that foster a service firm's ability to promptly respond and deliver improved services to quickly changing market needs. This study contributes theoretically to the B2B marketing literature, particularly regarding organizational capabilities in digital platform ecosystems, by presenting a framework and three propositions that illustrate service firms' pertinent organizational capabilities for long-term success. This study also offers avenues for further research as well as several implications for managers regarding building faster, more accurate service responses.

1. Introduction

The current turbulent times, characterized by rapid environmental, technological, socioeconomic, and geopolitical changes, are profoundly impacting societies and businesses, particularly service firms (World Economic Forum, 2022). To survive with new ways of working, and with increased uncertainty in business markets, service firms are undergoing major transformations to remain competitive and secure and to improve their market positions (Liu, Long, & Liu, 2023; Nanda & Narayandas, 2021). First, they are deploying digital multi-sided platforms (MSPs) to maintain interactions with their customers and to provide services remotely (Heikinheimo et al., 2024; Huang & Farboudi Jahromi, 2021; Liu, Chung, et al., 2023). Second, they are developing organizational capabilities to withstand market volatility and thrive in a rapidly evolving digital economy (Boh et al., 2023; Liu, Chung, et al., 2023).

MSPs are digital marketplaces that provide efficient matchmaking, relationship building, and transactions between two or more actors (Hagiu & Wright, 2015; Micallef et al., 2023; Wallbach et al., 2019) and are driven by cutting-edge technologies, such as artificial intelligence (AI) (Ameen et al., 2021; Huang & Rust, 2021). Examples of firms with digital marketplaces include Uber and eWork, the latter connecting

more than 130,000 professionals, such as software developers, to meet customers' needs remotely and globally with the help of an MSP. In both marketplaces, MSPs facilitate customer matching and drive service transactions via technologies deployed and integrated innovatively (see, e.g., Mancuso et al., 2024; Principato et al., 2023; Yuan et al., 2022).

In the business services context, MSPs transform traditional service value chains into platform-based virtual ecosystems of supply and demand, constituting a complex network of people, businesses, and technologies orchestrated by a platform owner (Heikinheimo et al., 2024; Hein et al., 2020; Perks et al., 2017). Leveraging MSPs with their technologies enables platform owners to reinforce these digital networks and strengthen their competitive advantages by increasing their resources and ability to react quickly to market changes (Floetgen et al., 2021; Liu, Chung, et al., 2023; Ricciotti, 2020). For service firms as platform owners, MSPs provide a structure to manage various internal and external experts and to deeply understand their customers' expertise needs, which is essential due to the nature of their service businesses (Heikinheimo et al., 2024; Nanda & Narayandas, 2021).

In addition to technologies, capabilities related to MSP ecosystems in a rapidly evolving digital economy help firms gain competitive advantages (Floetgen et al., 2021; Liu, Chung, et al., 2023; Rapaccini et al.,

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2020; Salmela et al., 2022). In MPSs, organizational capabilities, such as organizational agility (OA), organizational flexibility (OF), and organizational resilience (OR), help firms survive unexpected crises (Floetgen et al., 2021; Rapaccini et al., 2020; Salmela et al., 2022; Zia et al., 2023). Simultaneously, there is an incrementally growing need for further academic research focusing on the role of emerging technologies and platforms as organizational capability accelerators connected with firms' survival in turbulent times (Amankwah-Amoah et al., 2024). In our study, we follow Ramos et al. (2023) and Salmela et al. (2022) and consider agility and flexibility to be closely related but separate concepts. OA refers to a "firms' ability to continually be aware of and respond to sudden market changes" (Lee et al., 2015, p. 398). Conversely, OF is defined as "the degree to which an organization possesses a variety of actual and potential procedures to improve the controllability of the organization and environment" (Palanisamy, 2003, p. 84). Both agility and flexibility are linked in research with OR (Dubey et al., 2021; Yuan et al., 2022) as "enabling companies to adapt effectively to unexpected events, bounce back from crises, and foster future success" (Yuan et al., 2022, p. 87). OR builds long-term resistance to effectively overcome impending disruptive changes (Guo, Chen, et al., 2023; Ortiz-de-Mandojana & Bansal, 2016; Teixeira & Werther, 2013).

In this research, we adopt Singh et al.'s (2022) perception of the joint role of firms' agility, flexibility, and resilience in survival amid severe market disruption and highlight the importance of investigating these capabilities jointly. Moreover, we posit that MSP ecosystems offer distinctive opportunities for firms to enhance these capabilities for future success. Therefore, we argue that the capabilities must be explored separately, since each, in our view, has a unique embodiment within the MSP ecosystem. We state that OA constitutes short-term velocity and promptness with the help of AI and automation. OF is less time bound but manifests strategic utilization of flexible and reactive networks. OR indicates long-term competitiveness with future service offerings.

Earlier management research has shown that organizational capabilities are driven by dynamic capabilities characterized as sensing, seizing, and transforming (Teece, 2007; Teece et al., 2016) and constituting a set of high-level capabilities directed toward strategic change (Helfat & Raubitschek, 2018; Liu, Chung, et al., 2023; Salmela et al., 2022; Teece, 2007). Therefore, in this study, we examine how these indispensable organizational capabilities are constituted in the MSP ecosystem via dynamic capabilities theory (Teece et al., 2016). We utilize dynamic capabilities as a theoretical lens because the extant research has considered them a key factor in determining firms' survival in times of crisis (Gupta et al., 2024). Recent research has shifted focus to capabilities and seems to agree with their evident role in helping firms survive in turbulent markets (Amankwah-Amoah et al., 2024; Gupta et al., 2024; Singh et al., 2022).

Dynamic capabilities reflect a firm's ability to integrate, establish, and restructure internal and external resources to manage strategic change (Helfat, 2007; Matarazzo et al., 2021; Teece, 2007). Thus, dynamic capabilities allow firms to change how their business is conducted in an evolving ecosystem and to stretch beyond ongoing practices (Christofi et al., 2021; Zahoor et al., 2022; Zhou et al., 2019). Although strong dynamic capabilities are essential for enhancing OA (Ahmed et al., 2022; Liu, Chung, et al., 2023; Zahoor et al., 2022), OF (Ramos et al., 2023), or OR (Ali et al., 2022), studies linking OR, OA, and OF at the firm level are lacking in the platform ecosystem context (Conz & Magnani, 2020; Huang & Farboudi Jahromi, 2021). Notably, current organizational capability research lacks an in-depth understanding of platform-based ecosystems (Field et al., 2021; Floetgen et al., 2021), as earlier research has mainly focused on one capability at a time (Ahmed et al., 2022; Guo, Chen, et al., 2023; Liu, Chung, et al., 2023). As turbulence in the business market seems to be continuing due to rapid disruptive technologies, natural disasters, and geopolitical and economic insecurities, firms are building new strategies by utilizing technologies such as automation and AI to refine their ecosystems and gain

better long-term competitiveness (Amankwah-Amoah et al., 2024; Berger-de León et al., 2023). More academic understanding is needed to uncover how this is being implemented in real-life empirical contexts, specifically with the MSPs that firms utilize to maintain their competitiveness.

To address the gaps above, our research questions are as follows:

- How do the dynamic capabilities of service firms drive OA, OF, and OR in an MSP ecosystem?
- How do OA, flexibility, and resilience support service firms in an MSP ecosystem?

To address our research questions, we employ a qualitative research methodology (Gehman et al., 2018) with an abductive research approach (Dubois & Gadde, 2002). Between 2019 and 2022, we conducted an in-depth investigation focusing on 34 qualitative semi-structured interviews with CEOs, directors, and managers of business-to-business (B2B) service firms acting as platform owners orchestrating MSP ecosystems. This research contributes to the B2B marketing literature in four ways.

First, in terms of originality (Corley & Gioia, 2011), we respond to calls to empirically study the role of emerging technology-oriented capabilities in building firms' future success in the B2B context (Amankwah-Amoah et al., 2024) and how service ecosystems develop agility, flexibility, and resilience (Field et al., 2021; Floetgen et al., 2021; Liu, Chung, et al., 2023). We produce new theoretical knowledge on organizational capabilities in a platform-based B2B context, enhancing the current research (Guo, Chen, et al., 2023; Helfat & Raubitschek, 2018; Yuan et al., 2022). Second, in terms of incremental insights (Corley & Gioia, 2011), we enhance the existing conceptual understanding of three organizational capabilities—OA, OF, and OR—jointly used by service firms in digital platform ecosystems (Shashi et al., 2020; Teece et al., 2016) with our framework and three propositions. Third, we provide scientific utility (Corley & Gioia, 2011) by enhancing the current scant knowledge on dynamic capabilities as drivers of agility, flexibility, and resilience (Conz & Magnani, 2020; Teece et al., 2016). Fourth, our findings are useful in practice (Corley & Gioia, 2011) for managers of service firms, as they provide practical advice to increase capabilities to leverage the latest technology opportunities and resources to expedite service delivery and provide more accurate service responses to customer needs in volatile markets.

2. Theoretical background

To present the theoretical foundation of our study, we approach the literature from the perspectives of MSP ecosystems, organizational capabilities (OA, OF, and OR), and dynamic capabilities theory. Table 1 presents key studies and the positioning of our study compared to existing research. To our knowledge, no previous study has methodologically focused on investigating in depth how OA, OF, and OR are manifested in MSP ecosystems, as the majority of existing studies have focused on one capability and have used quantitative methods outside the MSP context (Ahmed et al., 2022; Liu, Chung, et al., 2023). Additionally, few studies have examined the link between dynamic capabilities and the three organizational capabilities that we have chosen (Aslam et al., 2018; Salmela et al., 2022; Teece et al., 2016), and none has done so within the platform-based B2B service industry context. Dynamic capabilities are a relevant but insufficient theoretical lens in determining firms' competitive advantages and survival in times of crisis (Eisenhardt & Martin, 2000; Gupta et al., 2024). Therefore, this study explores also other organizational capabilities in contemporary business environments.

2.1. Multi-sided platforms in the B2B service ecosystem

MSPs are gaining growing interest in advancing joint value creation,

Table 1
Positioning of our study.

Reference	Objectives of the study	Methodology	Dynamic capabilities	Multi-sided platform (MSP)	Organizational agility	Organizational flexibility	Organizational resilience	B2B services
Helfat & Raubitschek, 2018	Proposes three types of dynamic capabilities essential for platform leaders.	conceptual	✓	✓	-	-	-	-
Liu, Chung, et al., 2023	Reveals the role of digital strategic agility and digital platform capabilities in knowledge-intensive service firms.	quantitative	-	-	✓	-	-	✓
Floetgen et al., 2021	Shows that social and environmental practices with business sustainability enhance OR.	quantitative	-	-	-	✓	✓	-
Ahmed et al., 2022	Shows the importance of intellectual capital in enhancing OA for manufacturing SMEs via digital platform capability.	quantitative	✓	-	✓	✓	-	-
Teece et al., 2016	Proposes that strong dynamic capabilities are needed to promote OA.	conceptual	✓	-	✓	✓	-	-
Salmela et al., 2022	Conceptualizes digital agility based on previous research and publicly available cases.	conceptual	✓	-	✓	✓	-	-
Aslam et al., 2018	Reveals that market-sensing capability is a predecessor to agility and flexibility.	quantitative	✓	-	✓	✓	-	-
Guo, Chen, et al., 2023	Explores the interlinks between horizontal–vertical knowledge integration processes and resilience in multinational SMEs and the digital platform context in times of crisis.	quantitative	-	✓	✓	-	✓	-
Yuan et al., 2022	Investigates the mechanisms that produce OR in a platform-based sharing economy.	qualitative	-	✓	-	✓	✓	-
Schreieck et al., 2021	Describes the essential organizational capabilities for establishing platform ecosystems and co-creating and capturing value in these.	qualitative	-	✓	-	✓	-	✓
This study	Investigates how dynamic capabilities and MSPs enhance OA, OF, and OR in the B2B service trade.	qualitative	✓	✓	✓	✓	✓	✓

interaction, and even service transactions among B2B services (Huang & Farboudi Jahromi, 2021; Mancuso et al., 2024; Schreieck et al., 2021). For service firms, for whom experts producing services are the most important assets (Nanda & Narayandas, 2021) and internal professionals are often insufficient, MSPs provide a means to seek and manage expertise (Heikinheimo et al., 2024; Salmela et al., 2022). Existing research has witnessed MSPs transforming traditional modes of interaction, relationship building, and producing new types of dynamic networks (i.e., platform ecosystems consisting of at least the platform owner and two different actor groups, such as customers and complementary service providers) (Hagiu & Wright, 2015; Pagani & Pardo, 2017; Wallbach et al., 2019). In such networks, MSPs build value by matching various platform users, thus allowing for more efficient and direct service transactions among service producers and service buyers, even geographically distant ones (Liu, Chung, et al., 2023; Rana et al., 2023; Rasmussen & Petersen, 2017). This is because platform designs based on network-centric logic foster lateral collaboration among platform users (Parker et al., 2016; Wallbach et al., 2019).

Platform technology, such as AI (Alt, 2021; Huang & Rust, 2021; Kaplan & Haenlein, 2019), plays a crucial role in MSPs' operations by supporting the automation of the business processes along the customer journey, including service transaction and interaction processes (Alt, 2021; Ameen, Cheah, & Kumar, 2022; Ameen, Sharma, et al., 2022; Heikinheimo et al., 2024). Following Alt (2021) and Kaplan and Haenlein (2019), we define AI as “the ability embedded in a platform to accurately explicate and learn from internal and external data and accomplish specific tasks and goals through adaptation.” In this

research, AI is therefore considered an important element of MSP technology, fostering and supporting rapid and prompt service processes as well as advancing the utilization of data and data analytics (Alt, 2021). The evolution of MSP technology depends on the platform owner taking the initiative to establish and invest in the platform and attract various actors, such as complementary service providers and service buyers, to form a common ecosystem (Heikinheimo et al., 2024; Mancuso et al., 2024; Ricciotti, 2020; Teece, 2017). Therefore, platform owners play a critical role in designing and managing platform governance, laying the foundation for the activities, structures, and elements that characterize the MSP (Helfat & Raubitschek, 2018; Tian et al., 2021; Trabucchi & Buganza, 2020).

Besides platform owners, other actors may utilize MSPs' tangible resources, such as innovative new services and processes as well as data, and intangible resources, such as information and knowledge, to advance joint value creation in the platform ecosystem (He & Zhang, 2022; Perks et al., 2017; Ritala & Jovanovic, 2023). Therefore, MSPs may provide opportunities to improve operational efficiency and flexibility in parallel with enabling firms to respond rapidly to changes and provide long-term strategic responsiveness to constantly changing business markets (Ahmed et al., 2022; Guo, Chen, et al., 2023; Salmela et al., 2022; Wallbach et al., 2019). However, research focusing on MSPs as a means to produce more agile, efficient, and flexible business processes (e.g., Guo, Chen, et al., 2023; Yuan et al., 2022), is only starting to emerge, as can be seen in Table 1. Even rarer is research focusing on MSPs and their impact in the B2B service context (e.g., Schreieck et al., 2021). Regarding the recent turbulence in business markets, the benefits

of platforms in reinforcing firms' core capabilities, competitiveness, and survival therefore merit more academic attention.

2.2. Organizational agility, flexibility, and resilience in MSP ecosystems

OA enables firms to redirect resources quickly and promptly toward service value creation when internal and external circumstances, such as variations in demand and supply, require this (Christofi et al., 2021; Eckstein et al., 2015; Shashi et al., 2020; Swafford et al., 2006; Teece et al., 2016). OA is a time-driven concept connected to volatile business markets and illustrated by adjectives such as “rapid,” “prompt,” and “quick” (Salmela et al., 2022). Research has demonstrated the role of information technology in producing OA, including processes, communication technology, and knowledge (Chen & Chiang, 2011; Lee et al., 2015; Salmela et al., 2022; Sambamurthy et al., 2003).

Firms utilizing digital platforms with evolving technologies, such as AI, mobile internet, and big data, can quickly identify and react to market opportunities and may gain competitive advantages in turbulent times (Guo, Yin, & Liu, 2023; Lee et al., 2015). As shown in Table 1, research investigating OA in relation to digital platforms has emerged in recent years (Arias-Pérez et al., 2023; Guo, Chen, et al., 2023; Liu, Chung, et al., 2023; Ramos et al., 2023), although the majority has focused on agility in other-than-transactional platforms in the professional services context (Table 1).

The conceptualizations of OF are unclear (Christofi et al., 2021), and some researchers regard agility and flexibility as comparable concepts (see, e.g., Ahmed et al., 2022; Eckstein et al., 2015; Teece et al., 2016). We agree with Salmela et al. (2022), considering them to be closely related but separate platform-driven concepts. OF lacks the qualifier of being driven by sudden external shock (Salmela et al., 2022) but is related more to the firm's capability to be successful by responding and self-organizing operations to adapt to change (Palanisamy, 2003). However, the need for OF has been emphasized, as business environments are changing more dynamically (Byrd & Turner, 2001; Palanisamy, 2003; Phillips & Wright, 2009). Earlier research (Table 1) has examined OF in value chains by utilizing outsourcing and flexible labor arrangements (Aslam et al., 2018; Ramos et al., 2023). However, the focus in OF has recently shifted toward the strategic utilization of flexible and reactive networks, as many firms nowadays work as interconnected with various network actors (Ramos et al., 2023).

Empirical examples have demonstrated the platforms used to exchange market knowledge between retailers and suppliers to enhance flexibility (Chen & Chiang, 2011; Ravichandran, 2018). OF may involve the capacity to build platform architecture to allow innovative utilization of networks and resources (Hein et al., 2019; Ritala & Jovanovic, 2023; Salmela et al., 2022). In MSP ecosystems, platform architecture allows vertical and horizontal interconnectedness among various platform users, producing structural flexibility in resource enhancement (Shashi et al., 2020; Veile et al., 2022). By allowing networks and direct relationships among platform users to be harnessed innovatively, the available resources can be utilized flexibly according to demand, with service offerings widened (Cavallo et al., 2022; Hein et al., 2019; Ricciotti, 2020; Ritala & Jovanovic, 2023). However, more understanding is needed of how service firms can enhance OF in contemporary business environments, such as in MSP ecosystems, to enhance joint value creation and competitiveness, since such settings covered in the current organizational capability research are limited (Table 1).

The study of OR has gained momentum in recent years, driven by several global crises (Salmela et al., 2022). For instance, businesses have been striving to adapt their strategies to the consequences of COVID-19, geopolitical changes, or supply chain disruptions (Floetgen et al., 2021; Mysore et al., 2023; Sharma et al., 2020). Overall, the OR research is comprehensive, leading to various conceptualizations (Conz & Magnani, 2020; Floetgen et al., 2021; Guo, Chen, et al., 2023). Consequently, OR is viewed as a capability, a process, or an outcome (Conz & Magnani, 2020; Yuan et al., 2022) at the firm level. According to Ortiz-de-

Mandojana and Bansal (2016), OR can be seen as an underlying path-dependent capability to cope with dangers and problems. Since it is difficult to measure, its benefits are seen with a delay. Stemming from this, the recent research stream has connected OR to the shorter-term capabilities of flexibility (Sharma et al., 2020; Yuan et al., 2022) or agility (Guo, Chen, et al., 2023).

Despite the extensive research (Table 1) on OR, research connecting resilience in platforms is gaining rising interest but is currently more related to consumer business in mobility platforms (Floetgen et al., 2021), the platform-based sharing economy (Yuan et al., 2022), or manufacturing industries (Guo, Chen, et al., 2023). In such contexts, OR is driven by various factors, such as adaptive business models (Floetgen et al., 2021) and dynamic capabilities to absorb and leverage learning (Yuan et al., 2022). Nevertheless, to the best of our knowledge, no explorative studies focus on producing new knowledge on all three capabilities (OA, OF, and OR) in the MSP service ecosystem. Therefore, more research is needed to shed light on the capabilities necessary for service firms to succeed in turbulent business markets (Field et al., 2021; Floetgen et al., 2021; Hein et al., 2020).

2.3. Dynamic capabilities

Dynamic capabilities theory has been widely utilized in research to comprehend firms' strategic capabilities and their impact on competitive advantages (Teece, 2007). Dynamic capabilities have been regarded as important in highly volatile business markets, but the latest findings suggest that such capabilities are needed in all environmental conditions, such as organization-level routines and processes, to redirect firms' positions and strategic resources (Ahmed et al., 2022; Zhou et al., 2019). The modifier “dynamic” indicates abilities that help firms renew and restructure their resources and capabilities to boost their competitive advantages in various market situations (Gupta et al., 2024; Zia et al., 2023), making dynamic capabilities an appropriate theoretical lens for this study. Dynamic capabilities are organizational capabilities that can be classified as sensing, seizing, and transforming (Teece, 2007; Teece et al., 2016) and consist of management routines through which firms can redirect their resources (Eisenhardt & Martin, 2000). Such capabilities are essential for engagement with evolving platform ecosystems and their dynamics (Helfat & Raubitschek, 2018; Teece, 2018), but the research linking dynamic capabilities, especially in an MSP context, is limited, as presented in Table 1. Furthermore, Eisenhardt and Martin (2000) argued that dynamic capabilities are essential, but not sufficient, for firms to achieve a competitive advantage. Therefore, in this research, we focus on the role of dynamic capabilities in fostering organizational OA, OF, and OR capabilities in the MSP context for future competitiveness.

Sensing capability denotes a firm's ability to acquire knowledge and increase its understanding of new digital technologies, channels, and user behaviors (Badrinarayanan et al., 2022; Helfat & Raubitschek, 2018; Teece, 2017; Zia et al., 2023). Firms with high sensing capability likely continuously scan for new technological opportunities and disruptive threats, both internally and externally, and proactively create different scenarios based on the detected trends to understand their implications for the future (Badrinarayanan et al., 2022; Liu, Long, & Liu, 2023; Zahoor et al., 2022; Zia et al., 2023). Recent research has highlighted the importance of such sensing capabilities to focus on digital mindset building and scenario planning to survive the COVID-19 pandemic by harnessing digital platforms (Zia et al., 2023). For service firms facing pressures in turbulent business markets, strong sensing capabilities could enable them to recognize the possibilities of continuously evolving platform technologies, identify opportunities to engage with various actors, and attract them to such platforms by utilizing the latest cutting-edge technologies (Alt, 2021; Helfat & Raubitschek, 2018; Perks et al., 2017). For instance, accumulated real-time platform data could be utilized to analyze unexpected trends with the help of AI-driven predictive analytics to reduce human bias in sensing future turns

(Warner & Wäger, 2019).

Seizing capability develops as a continuum to sensing capability and advances the wider implementation of sensed technological opportunities via firms' practices, processes, and mobilization of resources for new ways of capturing value (Teece, 2007; Teece et al., 2016). Seizing capability involves implementing and scaling platform investments that are compatible with a firm's strategic objectives, evolving business models, and platform architecture (Hein et al., 2020; Teece, 2017). Furthermore, such capabilities enable service firms to plan the timing of investments to best support platform leverage within emerging MSP ecosystems (Helfat & Raubitschek, 2018; Teece, 2018; Yuan et al., 2022). Seizing capability also refers to the capability to change routines and allocate the internal and external resources and actions needed to extend platform networks and the services provided (Teece, 2007; Zahoor et al., 2022). Recent research has suggested that preventive actions driven by seizing capability, such as keeping resources in reserve in an MSP ecosystem and fostering employee training and multitasking to mitigate the negative consequences of market uncertainties, help firms survive with low-capacity utilization rates (Yuan et al., 2022). A key role in this is platform governance, which encompasses the design of platform structures and emphasizes the ease and flexibility of network enhancement (Hein et al., 2020; Perks et al., 2017; Teece, 2017). Overall, service firms' seizing capabilities strengthen their skills and competencies for planning and implementing complex chains of actions to leverage platforms as part of their ecosystems in various market situations.

Lastly, closely linked to sensing capability, *transforming capability* enables firms to balance continuity and continuous renewal with the reconfiguration of organizational resources and structures in the platform ecosystem (Cavallo et al., 2022; Liu, Long, & Liu, 2023; Teece, 2007). It could also involve regaining stability, increasing resources and capabilities, and reconfiguring routines as knowledge and learning accumulate (Ali et al., 2022; Teece, 2007; Zahoor et al., 2022). The capability to transform is essential for service firms to continuously challenge existing routines and adjust their operations (Cavallo et al., 2022; Zahoor et al., 2022). As service firms face unpredictable market situations with evolving platform ecosystems, transforming capability may help them overcome such changes (Teece, 2017; Zahoor et al., 2022). Recent research has shown the importance of transforming the capabilities of service firms to survive COVID-19 by transforming their business models and value creation activities by leveraging digital platforms (Gupta et al., 2024; Zia et al., 2023). Regarding platform leverage, transforming capabilities, such as enhancing continuous renewal, may be the utmost essence, as platform leverage is an ongoing process, and service firms need to keep up with the pace of technology development and competitive situations to continuously engage new actors and resources in platforms for the benefit of entire ecosystems.

To summarize, the lack of explorative research in the field of dynamic capabilities advancing the organizational capabilities of OA, OF, and OR in MSP ecosystems has led us to investigate this phenomenon from the perspective of service firms as platform owners. We posit that dynamic capabilities play a crucial role in building service firms' competitive advantages in both the short term (OA) and the long term (OF and OR). Therefore, it is necessary to investigate these capabilities jointly. As such capabilities are intended for strategic-level change and increasing competitive advantages in MSP ecosystems and volatile business markets, it is imperative to produce new in-depth knowledge of these capabilities and their linkage in supporting firms' survival.

3. Methodology

3.1. Research design and setting

To understand how firm-level dynamic capabilities foster OA, OF, and OR in an MSP ecosystem, we employed an abductive research approach (Dubois & Gadde, 2002, 2014). Such an approach is especially

suitable in situations in which theoretical frameworks exist, but it does not provide a full understanding of the empirical phenomenon (Dubois & Gadde, 2002; Tavory & Timmermans, 2014). Furthermore, the abductive research approach allowed us flexibility in the research design and reorientation of the analytical framework, while the researchers learned from the data to provide richer analytical explanations (Gehman et al., 2018).

Our firm-level investigation focused on service firms that deployed MSPs as digital places to make their business practices more efficient by digitizing the process of resource matching. The digital services in this research were provided by human experts, such as software coders. In addition to in-house experts, firms leveraged the external professionals of complementary service providers through MSPs. To address the limited B2B industry examples of MSPs (Heikinheimo et al., 2024; Micallef et al., 2023; Principato et al., 2023), we focused on B2B service firms as platform owners orchestrating MSP ecosystems.

3.2. Sampling

We employed purposeful sampling logic (Patton, 2015) to identify four B2B service firms that had introduced MSPs as a basis for their business environments. These firms were part of a large-scale research program focusing on the role of digital tools, AI, and automation in B2B sales. The firms were selected based on the following criteria: (1) during the start of this research, they had introduced MSPs as part of their B2B service sales; (2) they represented an industry known for utilizing digital solutions innovatively and thus providing a fruitful platform for investigating platform-driven capabilities; and (3) they were B2B service companies accessible to the researchers. These criteria allowed us to obtain a sample of 27 participants from four service firms (see Table 2). These participants were interviewed during two time periods: the first set of interviews was conducted between 2019 and 2020 and the second during 2022. The sample of 27 participants consisted of five women and 22 men, based on the researchers' assumption of their genders, and representing different levels of seniority and experience. The participants' industry experience varied between 3 and 29 years. The data collection and analysis are explained below.

3.3. Data collection and analysis

Data collection and analysis were conducted iteratively across four phases, as illustrated in Fig. 1. We begin by explaining the data collection process, then we describe the data analytical process. Regarding data collection, our research comprises two distinct datasets. In phase one, dataset 1 was gathered by interviewing 24 participants during 2019–2020, when the studied firms were at the beginning of their MSP adoption (see Table 2). Semi-structured interviews (see Brown et al., 2024) were conducted either face-to-face or via Microsoft Teams. The snowballing technique (Wilkinson & Young, 2004) was used to identify research participants within the sampled service firms by letting the firms' representatives suggest further interviewees. Phase one aimed to explore ways in which the service firms could enhance agility and flexibility within their service processes, particularly in the context of newly introduced MSP ecosystems.

In phase two, we revisited the service firms in November 2022 and conducted an additional 10 semi-structured interviews to confirm and enhance the initial understanding of OA, OF, and OR development (dataset 2). We collected dataset 2, as we wanted to give time for the MSP technologies and capabilities to develop in the service firms, since platform transformation takes time (Tian et al., 2021). Furthermore, the COVID-19 pandemic hit the service firms under investigation in spring 2020, hindering MSP adoption for a short while, as firms needed to adjust, for instance, into a fully remote working culture.

In phase three (see Fig. 1), we combined datasets 1 and 2 to confirm and enhance our understanding of organizational capabilities in the MSP context and their linkage with the dynamic capabilities of sensing,

Table 2
Illustration of the sample.

Interviewee ID	Position	Gender *)	Experience in years	Time of the interview (month/year)	Pages	Duration of the interview(s), h/min	Industry
#1	director, digital services	M	14	11/2019	23	1:31	Software design and development
#2	director, international business development	M	19	11/2019	25	1:38	Software design and development
#3	chief growth officer	F	29	11/2019	27	1:26	Software design and development
#4	product & development manager	M	5	11/2019	24	1:30	Software design and development
#5	capability accelerator	M	23	11/2019	29	1:29	Software design and development
#6	head of digital sales	M	20	11/2019	22	1:26	Software design and development
#7	sector lead, industry	F	9	11/2019	17	0:42	Software design and development
#8	sector lead, business	M	20	11/2019	19	0:52	Software design and development
#9	sector lead, society	M	27	11/2019	18	1:19	Software design and development
#10	growth marketing lead	F	8	11/2022	8	0:24	Software design and development
#11	head of sales	M	19	11/2019	27	1:41	Computer hardware and software consulting
#12	resource manager	M	10	11/2019	24	1:30	Computer hardware and software consulting
#13	chief information officer	M	20	11/2019	35	1:26	Computer hardware and software consulting
#14	business specialist	M	9	11/2019	25	1:30	Computer hardware and software consulting
#15	chief executive officer	M	25	11/2019	28	1:41	Computer hardware and software consulting
#16	chairman of the board	M	6	11/2019	21	1:12	Computer hardware and software consulting
#17	chief network officer	M	5	11/2022	11	0:29	Computer hardware and software consulting
#18	account manager	M	3	11/2022	16	0:32	Computer hardware and software consulting
#19	account manager	M	3	12/2019	51	1:29	Architectural services
#20	chief executive officer	M	29	12/2019	20	1:10	Architectural services
#21	customer success manager	M	9	11/2022	13	0:43	Architectural services
#22	R&D director	M	10	12/2019	14	0:54	Architectural services
#23	partner, innovation management	M	18	12/2019	17	0:53	Architectural services
#24	chief marketing officer	F	18	11/2022	21	1:24	Architectural services
#25	chief marketing officer	F	18	01/2020	9	0:29	Other management consulting
#26	account executive	F	7	01/2020	18	0:59	Other management consulting
#26	chief executive officer	M	19	01/2020	16	0:57	Other management consulting
#26	chief executive officer	M	19	11/2022	22	1:09	Other management consulting
#26	chief executive officer	M	19	11/2022	9	0:23	Other management consulting
#27	chief technical officer	M	7	01/2020	16	0:56	Other management consulting

* Based on the interpretation of the researchers.

seizing, and transforming (Teece, 2007). The entire dataset consisted of 34 interviews. Each interview lasted 22–102 min, with an average duration of 67 min. The data consisted of over 38 h of recordings and 666 pages of transcribed voice recordings for analysis (see Table 2).

Regarding the start of the data analysis, the transcribed interviews were first thoroughly read by the first and second authors to obtain an overall understanding of the data. In the first phase of the data analysis, dataset 1 was analyzed using the Gioia method (Gioia et al., 2013) to form an initial informant-centered understanding of the MSP's role in developing agility and flexibility in service processes and to identify first-order concepts of agility and flexibility in the MSP context (see Fig. 1). In the Gioia method, the first-order concepts capture participants' perspectives, followed by the second-order themes, and finally aggregate dimensions with the presentation of broader coherent topics (Gioia et al., 2013; Magnani & Gioia, 2023).

After a second round of interviews, the second phase of the analysis was conducted on dataset 2 to confirm and enhance our initial understanding of organizational capabilities and to identify initial second-order themes. Furthermore, we noticed an additional organizational

capability—resilience based on the informants' speech and feedback—which did not align with the characteristics of the other two capabilities. This led us to revisit the literature on organizational capabilities (see, e.g., Lee et al., 2015, for agility; Palasimany, 2003, for flexibility; and Yuan et al., 2022, for resilience).

In the third phase of the analysis, we merged datasets 1 and 2 into theory-driven coding and abductive analysis. This iterative coding confirmed and refined our initial first-order concepts, finally resulting in 75 codes (see Appendix A). The second-order themes were accordingly iterated, modified, and removed based on their relevance to the identified organizational capabilities of OA, OF, and OR, finally resulting in eight second-order themes. During this analytical phase, we identified a link between the higher-level dynamic capabilities of service firms and the organizational capabilities of OA, OF, and OR. Therefore, we again turned to the literature and employed a robust theory of dynamic capabilities (Teece, 2007), with its three categories of sensing, seizing, and transforming, as an analytical structure guiding our final analytical phase (Teece, 2007). Finally, we constructed aggregate dimensions of organizational capabilities (OA, OF, and OR) in an MSP ecosystem and

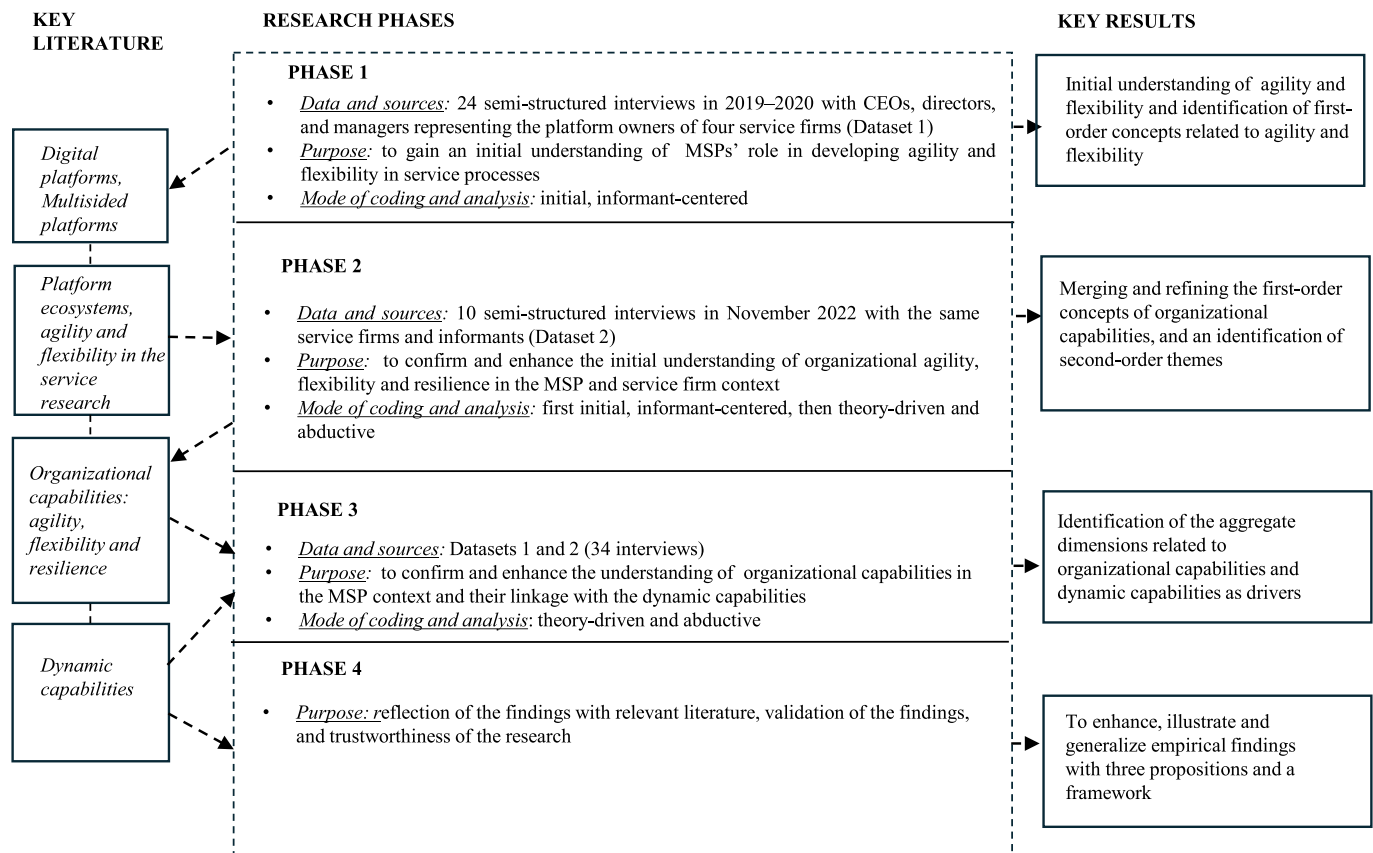


Fig. 1. Illustration of the data collection phases and analysis process.

linkages with dynamic capabilities as driving forces into those dimensions. An example of the final coding structure is presented in Table 3. In phase four, we reflected on the findings with the relevant literature and validated the findings and the trustworthiness of the research.

3.4. Trustworthiness of the study

To ensure the trustworthiness and quality of our research, we employed several well-established measures aligned with our qualitative research process and examples of recent research with abductive

Table 3
Example of a coding structure.

Sensing	Seizing	Transforming	First-order concepts	Second-order themes	Aggregate dimensions
X			- Data-based forecasting in demand - Harnessing real-time customer data on the platform - Allowing easy and open access to a platform database - Identifying AI's opportunities in sales analytics	Data-driven reactions to market needs	OA in the MSP context
X			- Utilizing AI in resource matching - Exploring the possibilities of platform technology - Doing technology and AI tool experiments	Accurate responses with automated resource matching	
	X		- Fostering boundary resources that enable integrations - Providing easy access to a service ecosystem - Low hierarchy with platform governance and between actors	Fostering an adaptable and evolving platform structure	OF in the MSP context
	X		- Building a common expert resource pool - Allocating new roles and resources for network development - Attracting foreign network resources	Effective use of common ecosystem resources	
	X		- Developing MSP-based processes - Reducing human work with AI-powered software robots and chatbots	Enhanced service range and processes	
		X	- Educating customers and a market for new platform-based services - Upscaling the platform and resources	Amplifying platform-based service ecosystem for expert trade	OR in the MSP context
		X	- Utilizing AI to scan and suggest competence gaps - Sharing data with various platform actors	Driving data-based development of future services	
		X	- Ability to assess and harness accumulated learning - Ability to continuously build partnerships and relationships - Constantly adapting to new markets and industries - Ability to overcome continuous improvements and technological challenges of the platform	Fostering continuous development and an open transformative mindset	

research logic, such as Sairanen et al. (2024). The careful data collection by the two first authors was followed by the establishment of a common research database with the help of ATLAS.ti software, enabling structured data management and iterative triangulation between the literature, data, and analysis following abductive reasoning logic (Dubois & Gadde, 2002; Farquhar et al., 2020). Interviewees were given the opportunity to provide commentaries on the research findings to enhance our analytical understanding and the validity of the findings. By collecting data from various professional informants over different time spans during the development of the MSPs, the authors increased the data and source triangulation. During our analysis, data saturation was reached when no new codes emerged from the data (Creswell, 2012).

During the coding process and data analysis, the two first authors maintained constant dialog with each other and with two other authors for reviewing the codes, findings, and analysis for researcher triangulation and, consequently, for increasing the trustworthiness of the research (Eriksson & Kovalainen, 2016). The university's research ethics protocols were carefully employed and explained to the research participants prior to the interviews and at the start of the interviews. For research transparency, the research process, including the analysis, was illustrated and documented in detail, following the criteria of Aguinis and Solarino (2019). The following section presents our findings, along with illustrative quotations from the interviews.

4. Findings

This study explores how dynamic capabilities foster OA, OF, and OR in MSP ecosystems in the B2B service context. Our findings highlight the impact of dynamic capabilities (sensing, seizing, and transforming) as higher-level management routines within MSP ecosystems, fostering service firms' organizational capabilities that enhance competitive advantages and survival in constantly changing business markets. A summary of the findings is presented in Table 4. In the upper section of Table 4, dynamic capabilities (sensing, seizing, and transforming) are presented from left to right as they appear in our data. The lower section of Table 4 focuses on presenting OA, OF, and OR as they appear in our data. Noteworthy was the linkage of sensing capability fostering agility, seizing capability of flexibility, and transforming capability of resilience. Based on our findings, strong sensing ability, such as the ability to detect and interpret data faster than competitors, helps platform owners build agility, such as making data-based decisions to provide accurate service responses. Strong seizing capability, such as the ability to harness network effects and optimal resource allocation for MSP ecosystem development, helps platform owners increase flexibility within the ecosystem through the effective use of common network resources, such as complimentary service providers in new markets. Lastly, strong transformation capability, such as a commitment to continuous platform development and learning, enables platform owners to enhance resilience, such as by reinforcing the new paradigm for service trade and building future data-based services. These capabilities are illustrated in more detail in the following subsections.

4.1. Sensing capability to foster organizational agility in the MSP context

MSPs, with their evolving technology, provide the context for service firms, as platform owners, to gain potential competitive advantages by enhancing agility in the service trade. The core of a service firm's value proposition is to quickly match the most suitable experts with various customer needs. Based on our findings, *sensing capability* is essential in enhancing OA in MSPs, as it pushes service firms to constantly scan for new technological opportunities and threats as well as obtain knowledge of the evolving technology to seek faster and more precise reactions to volatile market situations. Furthermore, sensing capability is essential for service firms to recognize the importance of data, identify new data sources, either internal or external, and predict market changes proactively. Consequently, OA manifests in our data with two elements: *data-*

Table 4
Summary of the findings.

	Sensing capability of a platform owner:	Seizing capability of a platform owner:	Transforming capability of a platform owner:
Dynamic capabilities based on managerial behavior	• Detecting internal and external data sources • Following the competitive situation • Identifying opportunities for and threats to platform technology • Obtaining knowledge of AI's opportunities • Constantly evaluating means for better resource matching	• Investing in a platform structure (people, tools, processes) that promotes flexibility • Allocating resources for the strategic development of the ecosystem • Seizing direct and indirect network effects • Promoting the use of common resources across country borders	• Continuous development of the platform and overcoming technological challenges • Continuously assessing own learning and the use of AI to scan for competence gaps • Assessing own and network resources
Organizational capabilities driven by sensing, seizing, transforming	Embodiment of organizational agility • Data-driven reactions to market needs • Accurate service response with automated resource-matching	Embodiment of organizational flexibility • Harnessing an adaptable and evolving platform structure • Effective use of common network resources • Enhanced service range and processes	Embodiment of organizational resilience • Amplifying a platform-based service ecosystem for expert trade • Driving data-based development of future services • Fostering continuous development and an open, transformative mindset

driven reactions to market needs and accurate service responses with automated resource matching.

Data-driven reactions in our findings are displayed as service firms' ability to harness real-time data in conducting faster data-based decisions in various areas, such as forecasting service demand in target markets. Data are collected from internal and external sources driven by platform owners' sensing capability. Internal data include platform data from past service projects, customer preferences, and the competences and soft skills of experts. External data encompass market trends and publicly available data sources related, for instance, to new programming languages, working methods (e.g., scrum), or search data of the most recent and sought-after skills. Data from various sources can be combined with AI and algorithms to produce various trend reports and forecast future market demand, appearing as enhanced OA, as illustrated in the quotes below:

[From the platform] we run reports, once a quarter, of the [skills and competences] that have been asked, and at the same time, we synthesize this to the knowledge base, which is then maintained and further developed.

(Interviewee 15)

The platform's database with smart functionalities and AI may detect the likelihood that the [potential lead] is close to a deal. ... AI is constantly learning about the behavior of current customers and about the time they became customers. [This information] is then utilized with a group of noncustomers' behavior data in the database to calculate the probabilities using algorithms.

(Interviewee 26)

Such data sources are essential for service firms, as platform owners, to detect whether changes in customer behavior, such as wishes to end ongoing projects, are short- or long-term phenomena in demand. For service providers, business success is dependent on high billing rates in the sense that experts' work time is efficiently allocated to hours that belong to billable customer work (Nanda & Narayandas, 2021). The ability to forecast future service demand, either decreases or peaks, is essential for service firms to be able to proactively cope with sudden shocks in the market. Therefore, the ability to harness real-time customer data on the platform is an important embodiment of OA in this context. Our findings also indicate service firms' active pursuit of identifying AI's opportunities in sales analytics to secure data-driven reactions to market changes and to secure high billable rates in any market circumstances, as illustrated in the following quotes:

If we see, for example, that there are going to be reductions in some client [projects]—which expert roles this affects, which expert resources we have free, looking forward, in [for instance] 2 months—then we should start selling that type of expertise to our clients right now. ... So this is [an example] of our data-based forecasting.

(Interviewee 1)

We have an AI-driven forecasting algorithm and feedback mechanisms that predict how much [an expert] would be able to bill for the customer for projects of different sizes. So if you see that [experts] are currently billing €120/h in some month-long project, and [the algorithm] predicts €25/h, then the [platform] would encourage [the experts] to improve their profiles.

(Interviewee 4)

OA also appears in our data by platform owners, allowing easy and open access to platform databases for other ecosystem actors. In this way, for instance, customers may utilize the data for planning their own projects and examining what kinds of competences are available. For service firms, such search data provide hints for possible future service needs and, again, help with demand forecasting, as illustrated by the following quote:

Now we have gone in the direction [with platform data usage] of being able to utilize it more proactively; customers could find those experts really quickly themselves [from the MSP with the support of AI] to plan their future projects.

(Interviewee 17)

OA was also displayed in our data as an *accurate response with automating resource matching*. Optimal matchmaking of experts or professional teams, such as software testers or software architects, to customer problems is at the core of the value proposition in the expert service trade. Whether utilizing a service firm's experts or those from complementary service providers, this matching relies on the platform's data on expert resources and previous projects. MSP, with its AI, facilitates this matchmaking by detecting required programming skills and technologies (e.g., Java or React), indicating suitable and available experts from the platform's database. AI, through language models that continuously learn, is utilized in resource matching. It can swiftly process a large number of platform databases to detect customer needs and basic information through specific questions. Automating resource matching enables quick reactions, but more importantly, it adds value by better matching to customer requests, benefiting not only service firms but also other users, such as customers and complementary service providers. This plays an increasingly prominent role in resource matching, progressively replacing slower human work as technology works more quickly, 24/7, and thus enhances OA:

At the core of [the platform] is simplified [matchmaking]; that is, the customer's need is identified precisely and understandably by utilizing the platform's knowledge base and AI. With the help of the

platform and AI, technical hard facts [concerning expert needs] can be tested very well. [Later,] expert profiles will also include softer values and ways of working.

(Interviewee 15)

We should constantly strive to get more things automated [in expert matching] to be able to trust AI, as AI uses much more data than one or 10 people.

(Interviewee 24)

Therefore, service firms, as platform owners, continually follow the development of platform technology, especially AI, driven by their sensing capability to provide better matches. This is highlighted in the following quotes:

I think that speed is the ultimate strength in this business. ... [The platform, with its database,] speeds up [the expert–customer matching] a lot because we don't have to ask the customer for all the basic information. We already know them [customers] since we have the data.

(Interviewee 11)

[The value of the platform for the ecosystem] is its transparency ... so that the pipeline is direct from the customer to the individual [expert]. At the start of the work, a customer getting the expert from the platform is faster than anywhere else. And, of course, customers get those talents that they want.

(Interviewee 12)

For automated resource matching, service firms conduct technology experiments to improve their performance in terms of reaction speed by leveraging platform attributes, as illustrated in these quotes. Besides AI, exploring other possibilities of platform technology is constantly being conducted to enhance agility driven by sensing capability. For instance, software robotics are utilized to sense human errors in work-time allocations, which are important to detect, as unreported work time would lower sales performance.

We [experiment with platform technology and AI all the time], and currently, we [automatically] check experts' hard technical facts, but we are [experimenting] with soft skills like team dynamics and [personality] types.

(Interviewee 15)

Our software robot checks our experts' work-time allocations. Because our sales are directly proportional to that, the experts remember to mark their work hours at the right place and at the right time. So [the software robot] sniffs everyone's work-time records and notifies people who lack the input or have done it perhaps in the wrong place.

(Interviewee 12)

4.2. Seizing capability to foster organizational flexibility in the MSP context

Based on our findings, the *seizing capability* of service firms is especially important in enhancing OF, as it encourages them to invest in platform structure, including people, technology, and processes, at the right time to enhance flexibility in the ecosystem. Service firms show seizing capability by allocating resources for the strategic development of the ecosystem, including scaling the identified technology opportunities and platform structure. Our findings indicate that seizing capability contributes to advancing OF, which is manifested through three key elements: *fostering an adaptable and evolving platform structure*, *effective use of common network resources*, and *enhanced service range and processes*.

Enhanced OF appeared in our findings, as platform owners *foster an adaptable and evolving platform structure* by enabling easy entrance to the

MSP ecosystem for various actors through its boundary resources. Through a platform's boundary resources, such as an application programming interface (API), integrations with other systems, such as customer relationship management systems, are possible and could be utilized, for instance, to target marketing messaging for the platform's customer engagement and to attract new platform actors. Additionally, integration into customers' procurement systems through APIs may provide enhanced business opportunities to service firms and other ecosystem actors. The following quotes illustrate the platform structure's relation to flexibility:

This whole [platform structure] is built on that adaptable and transparent model. This includes tools, processes, and people ... for [experts]. There is, for instance, the opportunity to see and choose the projects and clients that interest them. Choosing a customer assignment is self-directed in that sense.

(Interviewee 7)

Integration into different procurement systems will enable [us, with the platform,] to integrate [even more deeply] into the customer's processes, and then it will open the door for us to get bigger customers.

(Interviewee 12)

To foster an adaptable and evolving platform structure, a platform's AI is utilized to provide easy access to the MSP ecosystem, for instance, by utilizing a CV parser. This is a deep-learning AI solution that helps the new joining platform actor, such as an expert, by detecting the information from the CV input and automatically suggesting additions for missing details. This saves time and increases access to the platform due to easy entrance, which is especially relevant, for instance, to partners, as illustrated in this quote:

Various means have been taken to encourage [easy access to a platform], such as utilizing an AI parser structuring an existing CV and extracting the key information from there, which would speed up the creation of an expert profile and a CV [in the platform by the entering partner].

(Interviewee 17)

An adaptable and evolving platform structure is also manifested with a low hierarchy with platform governance and between actors additionally portraying OF. Such flexibility is bound to an adaptable platform structure that allows easy and direct collaboration among platform users in various locations. Therefore, any platform user may take the initiative to build relationships easily on the platform. Service firms are paying extra attention to ascertaining that such relationships may occur, as illustrated in the following quote:

The [platform's] user experience must be such that it doesn't feel [from the platform user's perspective] like you're at the end of a [long] chain but that the communication flows directly to the customer through the platform.

(Interviewee 12)

Driven by seizing capability, *the effective use of common network resources* as a form of OF is especially visible in the service firm's ability to build a common, evolving expert resource pool. This ability is crucial in the service trade, as it involves equalizing human resource capacity among various platform participants. Inviting complementary service providers onto the platform increases flexibility in the form of wider network resources, such as additional experts in a specific technology or competence area. Complementary service providers gain direct visibility of customers' needs on the platform to develop their service portfolios. As the boundaries blur between organizations and their experts in the MSP, expert resources are identified as common network resources, and service firms gain the flexibility to utilize these shared resources across various service requests. This flexibility can be employed not only to address demand spikes but also to offer a broader service response to

customer needs, as illustrated in the following quotes:

Instead of a single expert going to a customer assignment, the optimal solution for the customer would be to provide flexibility in the form of a dynamic team of five people whose specific knowledge is used for a certain amount in the customer project when necessary.

(Interviewee 5)

It's great when these new [service providers] come [on to our platform]. They're quite excited about it, and it makes the adaptation [to our system] easy, but the key thing, as I see it, is the know-how that they bring, and we can get their profiles [with project history] quickly visible; that is the best.

(Interviewee 1)

The effective use of common network resources is also manifested by service firms' allocation of new roles and resources for the benefit of the whole ecosystem, such as persons supporting complementary service providers with their concerns. As the platform's utilization is not dependent on time or place, it provides platform owners and other platform actors with the possibility of establishing network relationships with complementary service providers and customers in a distant market. This attraction of foreign network resources driven by seizing capability is illustrated as follows:

This year, we built a partner network [in Poland] like we have here. Sure, it's still significantly smaller, but it's already the size that you can run a business with it, so our focus has been strongly on extending the network [abroad] with the help of [the platform].

(Interviewee 15)

OF is also manifested by *enhanced service range and processes* through technology, such as utilizing AI-powered software robots and chatbots in service processes. The platform's technology, such as a dashboard with key statistics on service providers' expert sales, may provide valuable information to various platform actors and directly enable both service firms, as platform owners, and complementary service providers to display such services requested by customers. Developing MSP-based service processes culminates in the use of the advantages of platform technology, such as AI, in replacing previously human-driven tasks to enhance service processes, as illustrated in the following quotes:

This kind of optimization of our processes on MSP moves us forward and helps us stay on the course of development and be in the lead momentarily.

(Interviewee 2)

AI is [constantly learning] to understand related technologies [in the service delivery process]. For instance, in the case of [certain programming language skills], the AI may suggest a suitable expert [with close enough skills] based on the project history data [in the platform]. Therefore, the person looking for an expert does not need to understand [similar programming languages].

(Interviewee 11)

4.3. Transforming capability to foster organizational resilience in the MSP context

Based on our findings, transforming capability is connected to OR by fostering service firms' long-term competitive advantages and resistance to various market disruptions. Transforming capability manifests as service firms' ability to overcome technological challenges and is inevitable because the MSP ecosystem and competitive landscapes are constantly evolving. Service firms' ability to transform by constantly assessing their own learning compared to their strategic targets and continuously renewing themselves and common network resources can significantly enhance OR. Our findings illustrate enhanced OR through three essential elements: *amplifying platform-based service ecosystems for*

expert trade, driving data-based development of future services, and fostering continuous development and an open transformative mindset.

The adoption of MSPs in the service trade has marked a major transformation from traditional linear service value chains. By introducing MSPs into service sales processes and engaging users, service firms, as platform owners, have commenced a new paradigm for the service trade. With such strategic long-term change, service firms, driven by transforming capabilities, are constantly carrying out activities to *amplify the platform-based service ecosystem* with continuous renewal. Service firms are compelled to educate customers as service buyers and the market in general for the benefit of the MSP ecosystem. Compared to traditional value chains, such a platform-based ecosystem fosters OR through collaborative value creation and transparent and direct information flow. Our data revealed that service firms constantly reflect on gained learning and areas that need further improvement driven by transforming capability, as illustrated in the following quote:

This platform ecosystem is introduced to perform the current way more efficiently, and digitalization creates a new paradigm of doing things. When we do this, the whole organization has to change. So we are creating a new paradigm [for service selling] in this platform development.

(Interviewee 4)

Amplifying the platform-based service ecosystem for expert trade is portrayed in our data because the current state of MSP leverage is not considered satisfying enough. Service firms, as platform owners, are continuously preparing to upscale MSPs. This includes constant renewal by leveraging technological possibilities and considering ways to attract more platform users and integrate their expert pools on platforms.

We have goals, but there is always progress with those goals. We can get better all the time. And if we notice something isn't working, and we fail, then we change it. We have big goals in the direction we are going, but with smaller things, we progress through [trials and errors].

(Interviewee 6)

OR appeared in our data as an ability to *drive the data-based development of future services*. An MSP creates a common database of skills and competencies and enables both platform owners and other platform actors' easy access to information through its dashboard. This approach enhances OR by continually considering future services, leveraging a growing pool of network experts, and utilizing accumulated data. Monitoring competence gaps among in-house experts and the ecosystem network enables service firms to make informed decisions on service development and attract suitable complementary providers, thereby enhancing long-term OR in the service trade. A platform's AI is crucial in driving data-based service development, as it scans accumulated data, identifies internal competence gaps, and compares them to external data on technologies and current skill needs. It generates this information in various forms, such as segmented into different markets, as explained in the following quotes:

[On the platform,] we can manage much better not only our own know-how but also our network's know-how. That is, we [and other platform users] see much better what skills we have [to develop future services].

(Interviewee 11)

Let's take Bob as an example. For some reason, Bob hasn't received a single customer assignment in the last 6 months, so then, [the platform's] AI can identify that Mary has taken all those assignments and compare the knowledge gap between Bob and Mary. Mary seems to have some scripting skills that Bob is missing. Therefore, pointing out [missing] skills and training needs at an individual level is valuable, but on a larger scale, we can predict trends such as identifying

repeatedly missing technologies and rising demand, which indicates areas for future improvements.

(Interviewee 15)

Data-based development of future services is also displayed, with the data being transparently shared with other platform actors. For instance, customers may utilize the data to plan their own software projects by seeing available expertise and providing service firms with ideas about future service needs. Likewise, complementary service providers may utilize this information to develop their own skills to match future technical skill requirements and trends, as illustrated in the following quote:

The data that the platform provides has hugely increased our partners' knowledge of complicated customer requirements, processes, and what they actually do and why they do things the way they do.

(Interviewee 23)

OR is additionally portrayed as *fostering continuous development and an open, transformative mindset* driven by transforming capability. This relates to the ability to foster an innovative mindset to foster and strengthen the MSP ecosystem and to see the possibilities that such an ecosystem may generate, as illustrated by the following quote:

Our aim is to provide a platform to manage service competencies and utilize the same platform to resource [experts and teams] for different projects and find those world-class experts who are there in the ecosystem. And why doesn't some local industrial company think, "Heck, we don't have a job for our engineers right now, so let's put them up for sale in the ecosystem," in which case there's no need to fire anyone, but instead, they'll get a chance to do a consulting gig.

(Interviewee 4)

Fostering continuous development and an open transformative mindset are manifested with the ability to assess and harness accumulated learning and are therefore an important form of OR as an ecosystem and as its technology constantly evolves, as illustrated in the following quote:

In practice, when we talk about the comprehensive move [into service practices] and the management of our network on the platform, we have taken on more responsibility for platform orchestration, even though it is shared. This responsibility is no longer concentrated in a single person.

(Interviewee 9)

Fostering continuous development and an open transformative mindset are also displayed, as the service ecosystem is based on the ability to engage various actors in the platform. Therefore, the ability of a service firm to continuously build partnerships and relationships is at the heart of the success of the platform. This is also manifested by the service firms' ability to constantly adapt to new markets and industries, whereby the platform can provide benefits in matching the proper service experts with customer needs, as illustrated in the following quote:

Of course, it depends on what the actual business potential is [in different service industries], but when the use of AI and algorithms is successful, ... and as the technology is built based on the assumption that the platform is not industry dependent, if we look at whatever [service industry, the same problem [of matching proper experts to customers' problems] will come up there too.

(Interviewee 15)

5. Discussion

This study explores how the dynamic capabilities of service firms foster OA, OF, and OR in MSP ecosystems. Our model (Fig. 2) demonstrates the enhancement of organizational capabilities (OA, OF, and OR) through which dynamic capabilities (sensing, seizing, and transforming)

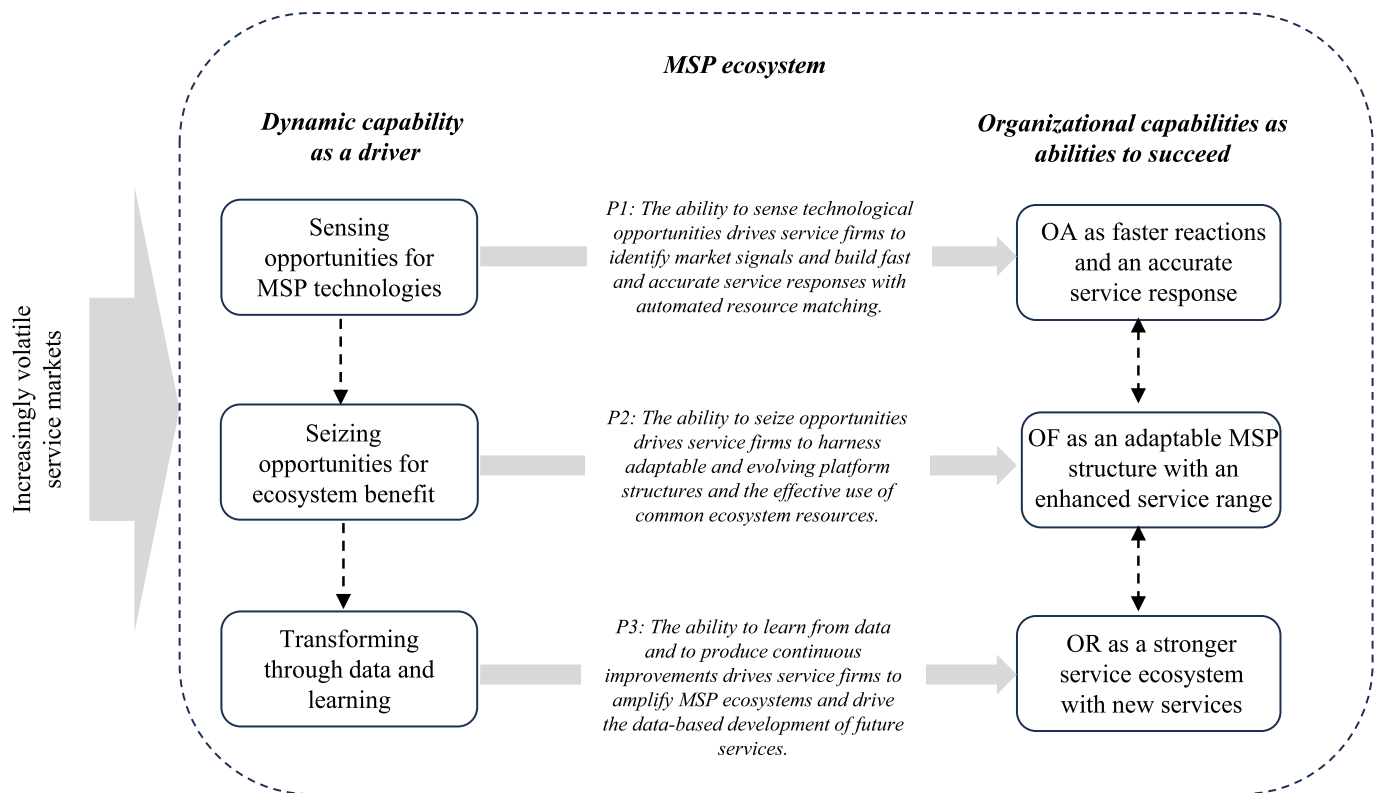


Fig. 2. Model of dynamic capabilities fostering organizational agility (OA), organizational flexibility (OF), and organizational resilience (OR).

built as a continuum on top of each other (Teece, 2007) serve as drivers in the MSP ecosystem. We also propose three propositions to show how each dynamic capability is connected to the organizational capabilities of OA, OF, and OR. The following discussion delves more deeply into the evolution of each capability, emphasizing its interconnectedness for superior competitive advantages and long-term preparedness for future disruptions.

5.1. Service firms' sensing capability driving organizational agility

Our findings indicate that service firms have introduced MSPs to provide the basis for building a service ecosystem that enables rapid and accurate service responses to fluctuating market demands. MSPs' accumulation of data, combined with external data sources, drives service firms, as platform owners, to carry out data-driven reactions to market needs and enhance AI-driven forecasting based on market signals. Furthermore, AI is utilized in MSPs to provide automated resource matching and to generate accurate service responses to customer assignments.

These embodiments of OA are vital for service firms to enhance their short-term competitive advantages via fast and precise actions. Previous research has shown OA's role in helping firms survive better through turbulent and volatile surroundings (Ahmed et al., 2022; Guo, Yin, & Liu, 2023; Liu, Chung, et al., 2023; Teece et al., 2016). However, in this previous research, OA is a concept comparable to OF, while our findings suggest that OA is a separate, largely platform technology-driven organizational capability in an MSP ecosystem. Based on our findings, OA with faster reactions and accurate responses is displayed as data-driven reactions and automated resource matching efficiently handling customer needs, not only improving speed but also enhancing service quality through its ability to process larger datasets faster and with fewer errors than humans.

Our findings extend the understanding of OA in the digital platform context (Ahmed et al., 2022; Liu, Chung, et al., 2023; Sambamurthy

et al., 2003) by highlighting its evolution, driven by service firms' sensing of technology opportunity capability. Sensing capability, the importance of which has been identified in previous research (Badrinarayanan et al., 2022; Guo, Yin, & Liu, 2023; Gupta et al., 2024; Teece, 2017), empowers service firms to identify the evolving technological opportunities of platforms, including data and AI. This high-level capability (Fig. 2) acts as a driver for OA, contributing to faster reaction speeds and accurate service responses to unexpected market needs. Based on this, our first proposition is formulated as follows:

P1. The ability to sense technological opportunities drives service firms to identify market signals and build fast and accurate service responses with automated resource matching.

5.2. Service firms' seizing capability driving organizational flexibility

Based on our findings, service firms, as platform owners, design MSP structures that enhance accessibility and extend reach not only to the surrounding network but also to complementary service providers and customers to form service ecosystems with enhanced service range and open, transparent processes. This structure facilitates the knowledge and skills of various experts as well as direct and multilevel collaborations between ecosystem participants, allowing open utilization of network resources for customer assignments.

A platform's structure supports collaborations and relationship building irrespective of time or location, boosting the service firm's capacity to expand networks internationally, thereby enhancing the adaptable, evolving service range. Given the constant evolution of MSPs and the networks built into their ecosystems, OF dynamically evolves over time. As such, we enhance the existing research (Byrd & Turner, 2001; Salmela et al., 2022; Teece et al., 2016) by showing how flexibility in the platform context provides service firms with a unique competitive advantage, as OF is related to service firms' ability to foster adaptable MSP structures, enhance service processes, and harness common platform resources effectively.

Building on previous research highlighting how seizing capability drives the implementation of identified technological opportunities (Teece, 2007; Teece et al., 2016; Zahoor et al., 2022), our study shows that within dynamic capabilities, a service firm's seizing capability acts as an enabler for OF. This capability allows service firms, as platform owners, to harness shared network resources in MSPs effectively (Fig. 2), expand service offerings, and foster adaptable MSP structures. Seizing capability relates to service firms' willingness to continuously invest and wisely allocate resources to implement identified technological opportunities innovatively in platform-based processes and to leverage network effects to add additional sides to platforms (Sambamurthy et al., 2003; Teece, 2007; Wallbach et al., 2019; Zia et al., 2023). Therefore, we present our second proposition:

P2. The ability to seize opportunities drives service firms to harness adaptable and evolving platform structures and the effective use of common ecosystem resources.

5.3. Service firms' transforming capability driving organizational resilience

Lastly, transforming capability allows service firms, as platform owners, to learn from accumulated data, continuously advancing the MSP ecosystem by portraying OR. With the help of platform data, transforming capability serves as a driver for the development of OR in the long term, as indicated by our results, through the ongoing expansion of MSP ecosystems and future data-based services. We therefore enhance the existing knowledge of OR related to platforms (Ali et al., 2022; Boh et al., 2023; Guo, Chen, et al., 2023; Liu, Long, & Liu, 2023) by showing resilience as organizational capability in the platform-based service ecosystem, since the current literature has focused more on digital resilience (Boh et al., 2023) or other industries (Ali et al., 2022; Guo, Chen, et al., 2023). Using cutting-edge technology, such as AI, enables the monitoring of knowledge and competence gaps in MSPs, prompting actions and development to strengthen existing service portfolios.

Driven by the transforming capability emphasized by earlier strands of the literature (Ali et al., 2022; Teece, 2007; Zahoor et al., 2022), the ability to harness accumulated data in assessing one's own and network resources in the ecosystem, as well as learning, enhances OR as a long-term improvement of competitive advantages in volatile markets. This ability, rooted in detecting changes in technologies or demand patterns, fosters the continuous development of service offerings, building a foundation for long-term preparedness against upcoming challenges. OR-evolving MSPs and accumulated learning are capabilities that develop over time. Therefore, we suggest the following as our third proposition:

P3. The ability to learn from data and to produce continuous improvements drives service firms to amplify MSP ecosystems and drive the data-based development of future services.

6. Conclusions

This study focused on producing novel insights into organizational capabilities within the B2B service context in MSP ecosystems. Drawing on an exploratory abductive study of four B2B service firms as platform owners orchestrating and MSP ecosystems, we found that each organizational capability (agility, flexibility, and resilience) is needed jointly to provide service firms with short- (OA) and long-term (OF, OR) competitive advantages and improved readiness to withstand future disruptions. We posited that technology, mostly AI and automation, produces speed and accuracy in service responses, outperforming human behavior as OA. OF is a less time-bound, separate concept of OA and culminates in the optimal utilization of adaptive platform structure and common network resources. In particular, OR builds long-term responsiveness to and competitiveness with data-based service

offerings and the reinforcement of the platform-based service selling paradigm. Additionally, the development of these capabilities requires the strong sensing, seizing, and transforming capabilities of the platform owner.

6.1. Theoretical contributions

In this research, we applied the theoretical perspective of dynamic capabilities within B2B service firms to investigate OA, OF, and OR in the MSP ecosystem. This study makes three key contributions to the B2B marketing literature by producing new knowledge about organizational capabilities in digital platform ecosystems.

First, we contribute to the existing B2B marketing literature in terms of originality (Corley & Gioia, 2011). Our research provides novel and original theoretical knowledge on B2B service firms building agility, flexibility, and resilience in the contemporary business environment (i. e., a digital multi-sided platform ecosystem). By focusing jointly on these three organizational capabilities, we respond to calls to study the role of emerging technology-oriented capabilities in building firms' future success in the B2B context (Amankwah-Amoah et al., 2024). Additionally, service research (Field et al., 2021; Floetgen et al., 2021; Liu, Chung, et al., 2023) has called for a new understanding of how service ecosystems develop agility, flexibility, and resilience with qualitative data. Although the importance of all three capabilities jointly helping firms survive in times of crisis, such as COVID-19, has been highlighted by Singh et al. (2022), we are the first to examine these capabilities as providing fast and accurate service responses to volatile market demand as well as long-term resistance and enhanced service range as preparedness for future crises.

Second, we offer a framework and three propositions to demonstrate the new content of organizational capabilities in the MSP ecosystem. OA is driven by platform technology, mostly by AI, data-based decisions, and automated tasks, to generate short-term velocity and accuracy in service responses outperforming human behavior. OF is less time bound but manifests strategic utilization of flexible and reactive networks, and OR shows long-term competitiveness with a future service offering and platform-based service selling paradigm. These incremental insights (Corley & Gioia, 2011) enhance the existing organizational capability research (Guo, Chen, et al., 2023; Liu, Long, & Liu, 2023; Salmela et al., 2022). Additionally, we confirm that the difference between OA and OF is that they bring different competitive advantages for firms' survival in turbulent times and, as such, contribute to the literature on agility and flexibility, in which the definitions have been somewhat unclear (see Christofi et al., 2021; Salmela et al., 2022; Teece et al., 2016).

Third, we bring scientific utility (Corley & Gioia, 2011) to B2B marketing research on capabilities. We build a theoretical bridge between dynamic capabilities theory, which has recently highlighted their role as defining firms' success or failure in times of crisis (Gupta et al., 2024; Zia et al., 2023), and agility, flexibility, and resilience research that has mainly focused on one capability on time (Christofi et al., 2021; Conz & Magnani, 2020; Eckstein et al., 2015; Guo, Chen, et al., 2023; Ramos et al., 2023; Shashi et al., 2020; Swafford et al., 2006; Teece et al., 2016; Yuan et al., 2022). We show that OA, OF, and OR are not purely enabled by MSPs but still require dynamic capabilities as management behaviors of service firms to drive the emergence of such capabilities. Therefore, our research enhances the current understanding of the role of dynamic capabilities as drivers (Conz & Magnani, 2020; Teece et al., 2016; Zahoor et al., 2022) by highlighting their importance in helping firms sense the opportunities of MSPs and AI by seizing the platform structure for ecosystem benefit and transforming the service selling paradigm through data and learning.

6.2. Managerial contributions and implications for policymakers

Our results are useful in practice (Corley & Gioia, 2011) for managers and policymakers, as this study provides novel insights into

organizational capabilities within the MSP ecosystem. Specifically, we demonstrate how these capabilities can contribute to both short- and long-term competitive advantages as well as enhance preparedness in volatile market conditions. First, we recommend that managers of service firms reinforce their sensing capabilities and experiment with various technological options, including AI and automation, to keep up with technological developments and reinforce their agility to react quickly to market disturbances while focusing on tasks in which technology can outperform manual efforts. Furthermore, AI can be used to trigger human-driven actions, such as data-based forecasting of demand and accelerating service delivery. Operational efficiency and automation can bolster short-term competitive advantages by enabling swift responses to changing market needs. AI's capacity to process diverse datasets also enhances the ability of service firms, as platform owners, to respond more accurately to customers' needs. As demonstrated in this study, this includes improving matchmaking in the platform ecosystem.

Second, we recommend that managers of service firms use their seizing capability by focusing on designing and investing in business structures (people, tools, and processes) that promote flexibility. In this research, this has meant multi-sided platform governance and structure that supports easy network entry, direct collaboration, and transparent information sharing. These activities play a critical role in attracting suitable complementary service providers to platforms along with their expert resources, thereby enhancing the knowledge and service offerings available in the MSP ecosystem. As shown in our study, an adaptable platform structure may even enable the extension of a network across country borders, providing service firms with a wider service range to meet diverse customer needs, thereby enhancing flexibility as an effective use of common network resources in various market situations.

Third, sustaining the continuous renewal of service firms and fostering the growth of the platform-based service trade can establish long-term resilience by transforming capability and better preparing for impending crises. Furthermore, drawing on the accumulated knowledge of platform technology usage, such as accumulated data and addressing challenges with an open and transformative mindset, can fortify service firms' responsiveness in navigating future crises.

For policymakers, we show that digital platforms are increasingly changing the business paradigms of B2B firms. Today's business leaders are also policymakers because platform companies educate markets and actors and leave strong footprints on how societies and individuals think, communicate, and operate. Policymakers, with their support for investment, education, and training, should especially encourage traditional firms, such as the service firms in this research, to increase their capabilities to harness the benefits of technologies such as easily accessible ecosystems to keep up with the competitive business environment and secure local job opportunities. Because modern humanity is characterized by rapid technological changes combined with turbulent times and environmental, socioeconomic, and geopolitical changes, agility, flexibility, and resilience should also be encouraged as organization-level capabilities at the levels of society and individuals. Policymakers also have the responsibility to consider the negative effects of increased technology with speed, availability, and connectivity and consider the dark sides, for instance, with legislative and regulative perspectives for building better societies and long-term resistance to future disruptions.

6.3. Limitations and future research

Our study has several limitations that offer potential for future research directions. First, we based our findings on interviews from 2019 to 2022 with 27 key informants from B2B service firms as MSP owners. The embodiment of OA, OF, and OR in the contemporary platform environment in one industry sector provides a starting point for several future research investigations. First, our focus on service firms as platform owners excludes perspectives from other actors, such as

customers and complementary service providers. Therefore, future research should examine multiple perspectives on the organizational capability-building process.

Second, our qualitative and abductive research provides a holistic view of OA, OF, and OR in an MSP ecosystem driven by dynamic capabilities. However, as a next step, it would be interesting to measure and test our model and propositions with a larger sample of firms. Third, as platform transformation takes time, and in our case, the firms have succeeded in leveling up their organizational capabilities in the MSP ecosystem, not all MSP ecosystems are successful and survive in the long run. Therefore, it would be interesting to reveal the development patterns and underlying mechanisms of unsuccessful B2B platform ecosystems that failed to increase their capabilities to resist competitive forces or disruptions in the industry. Fourth, we concentrated on MSPs in one B2B industry providing, for instance, IT services. Even though these firms were traditional firms that introduced MSPs as part of selling their services, they may have had better readiness with dynamic capabilities and agility. Therefore, future research should examine such organizational capabilities in various industries, such as public services, disrupted by platforms.

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CRediT authorship contribution statement

Heikinheimo Minna: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hautamäki Pia:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Julkunen Saara:** Writing – review & editing, Validation, Supervision, Software, Resources, Funding acquisition, Formal analysis, Conceptualization. **Koponen Jonna:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.indmarman.2025.01.006>.

Data availability

The data that has been used is confidential.

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